

Study Questions: From Idea to Intervention

Analytical Questions

1. Active Inference distinguishes between perceptual inference (updating the model to match the world) and active inference (changing the world to match the model). Explain how prototyping involves both types of inference simultaneously. When does the inventor change the world, and when do they update their model?
2. The Wright Brothers decomposed powered flight into sub-hypotheses and tested each with a targeted prototype. Contrast this approach with Samuel Langley's strategy of building a full-scale powered aircraft directly. Using Active Inference concepts (expected free energy, epistemic value, pragmatic value), explain why the Wrights' incremental approach was more efficient.
3. Explain the concept of a "minimum viable prototype" (MVP) using Active Inference terminology. Why should the MVP test the riskiest assumption rather than the easiest one? What happens to expected free energy when the riskiest assumption is resolved?
4. James Dyson built 5,127 prototypes of his cyclonic vacuum cleaner. Analyze this iterative process using the perception-action loop framework. Was each prototype a separate experiment, or were they stages of a single extended experiment? What role did prediction error play at each iteration?
5. Chuck Hull's invention of stereolithography demonstrates how action can generate entirely new problem spaces. Explain how building the first 3D-printed cup created prediction errors (layer bonding, resin shrinkage, surface finish) that Hull's mental model had not anticipated. Why could these errors only have been discovered through action rather than through further mental simulation?
6. Active Inference proposes that actions should minimize expected free energy, which balances epistemic value (uncertainty reduction) and pragmatic value (goal achievement). Early prototypes should prioritize epistemic value. When should an

inventor shift from epistemic to pragmatic prototyping? What signals indicate this transition?

7. The perception-action loop operates at different timescales: seconds (tool adjustment), days (prototype testing), months (design iteration), years (product evolution). How do these nested loops interact? Can learning at one timescale conflict with learning at another? Give an invention-specific example.
8. Explain why informative failure is more valuable than ambiguous success. Design a prototype scenario where a successful test provides less information than a failed test. Under what conditions is "it worked" actually less useful than "it broke at this specific point"?
9. The concept of "acting under uncertainty" suggests that inventors should build when mental simulation reaches its limits. But how does an inventor know when they have reached this limit? What cognitive signals indicate that further thinking will not reduce uncertainty, and only action can?
10. Rapid prototyping technologies (3D printing, laser cutting, breadboarding, code sandboxes) have compressed the perception-action loop from weeks to hours. Analyze the impact of this compression on the inventive process. Does faster prototyping always lead to better inventions? What are the risks of making the loop too fast?

Applied Questions

1. Identify the three riskiest assumptions in your invention idea. For each, design a minimum viable prototype that would test it. Specify: what you would build, what you would measure, what result would confirm the assumption, and what result would disconfirm it.
2. Think about a time when you built something (any domain — cooking, crafting, coding, assembling furniture, writing) and discovered something unexpected through the building process. Describe what you learned and why you could not have learned it without acting on materials.

3. Design a prototype testing plan for your invention with three iterations. Each iteration should test a different assumption, building on the knowledge gained from the previous one. Show how the perception-action loop connects the iterations.
4. Your prototype failed — the mechanism does not work as you predicted. Walk through the Active Inference process for handling this failure: (a) identify the specific prediction error, (b) determine which part of your generative model was wrong, (c) update your model, and (d) design the next prototype based on the updated model.
5. Compare the cost-information ratio of three prototyping approaches for your invention: (a) a paper sketch, (b) a physical mockup from cheap materials, and (c) a functional prototype from real materials. Which approach gives the most information per dollar and per hour invested at the current stage of your project?
6. Reflect on your own "build threshold" — the point at which you shift from thinking to building. Do you tend to think too long before building, or build too quickly before thinking? What evidence supports your self-assessment? How would adjusting your threshold change your inventive process?
7. Design a prototype that is specifically intended to fail — one where the failure will be maximally informative about a critical variable. What variable are you testing? What range of failure conditions will you explore? How will each failure mode update your model differently?
8. Consider the role of tools in the perception-action loop. What tools do you currently have access to for prototyping? How do these tools shape what you can test? If you had access to one additional tool (specific tool, not "better tools in general"), what new prototype experiments would it enable?
9. Your invention has both a physical component and a user interaction component. Should you prototype the physical mechanism first or the user interaction first? Use the concept of expected free energy (balancing epistemic and pragmatic value) to justify your choice.
10. Write a "prototype learning journal" entry for a real or hypothetical prototype test. Include: the prediction being tested, the prototype design, the test procedure, the results,

the prediction errors identified, the model updates made, and the plan for the next iteration. This journal entry should read like a scientific experiment report, adapted for invention.